NPaxis Platform

Backend Development — Project Report

One Month Execution Plan | 40 Hours | 20 Development Days

Prepared by: Project Manager & Backend Lead

Field	Details
Company	Digitalearn Solution Pvt. Ltd.
Product	NPaxis Platform
Role	Project Manager / Backend Designer
Total Duration	1 Month (20 Working Days)
Total Hours	40 Hours (2 hrs/day)
Report Date	March 2026

1. Executive Summary

NPaxis is a governed, role-based digital platform developed by Digitalearn Solution Pvt. Ltd. It connects nursing/medical students with verified preceptors through a secure, monetized, and compliance-driven ecosystem.

This report documents the complete one-month backend development plan, covering all 40 hours of engineering effort distributed across 20 working days. The backend is designed using Node.js, Firebase (Firestore, Auth, Cloud Functions), and Stripe for payment processing.

As the Project Manager and Backend Architect, this plan encompasses authentication, role-based access control, preceptor verification workflows, premium subscription monetization, contact reveal gating, analytics tracking, and admin governance APIs — all built on a secure, scalable, serverless-first architecture.

Key Metrics at a Glance:

Total Hours	Working Days	API Endpoints	Modules
40 Hours	20 Days	25 Endpoints	8 Modules

2. Platform Architecture Overview

The NPaxis backend follows a modular, serverless-first REST API architecture. The system is structured around eight core backend modules, each responsible for a distinct domain:

- Authentication & RBAC Module — Firebase Auth + role-based middleware
- Student Profile Module — profile management, bookmarks, search
- Preceptor Profile Module — profile creation, verification workflow
- Admin Governance Module — approval queue, user management, platform stats
- Payment & Subscription Module — Stripe Checkout, webhooks, billing portal
- Contact Reveal Module — premium-gated contact access with audit trail
- Inquiry & Notification Module — structured inquiry flow + email dispatch
- Analytics & Reporting Module — event tracking, preceptor stats, admin KPIs

2.1 Technology Stack

Layer	Technology	Purpose
Runtime	Node.js + Express.js	Backend server & REST API framework
Database	Firebase Firestore	NoSQL cloud database for all collections
Auth	Firebase Authentication	JWT-based auth, email verification
Storage	Firebase Cloud Storage	License document file uploads
Functions	Firebase Cloud Functions	Serverless triggers & email notifications
Payments	Stripe API	Subscription checkout, webhooks, billing portal
Email	SendGrid / Nodemailer	Transactional email delivery
Security	Helmet.js + Rate Limiter	HTTP headers & request rate protection
Docs	Swagger / OpenAPI 3.0	Auto-generated API documentation

3. Month-Long Development Plan

The 40-hour development plan is distributed across four weeks of five working days each. Each day represents a 2-hour focused development session. The plan is ordered by dependency: infrastructure and auth first, then feature modules, then integrations and testing.

Week 1 (Day 1 – Day 7) — Project Setup, Architecture & Auth System

Day	Hours	Task	Scope & Description	Status
Day 1	4 hrs	**Project Initialization**	Firebase project setup, Firestore config, environment variables, repo structure, CI/CD pipeline init	Planned
Day 2	4 hrs	**Authentication Module**	Firebase Auth integration: sign-up, login, email verification, JWT token handling, role assignment on registration	Planned
Day 3	4 hrs	**RBAC Middleware**	Role-Based Access Control middleware for Student / Preceptor / Admin, route protection, auth guard functions	Planned
Day 4	4 hrs	**Database Schema Design**	Firestore collections: Users, Students, Preceptors, Payments, Analytics_Events, Preceptor_Contacts — index setup	Planned
Day 5	4 hrs	**Student Profile APIs**	POST /students/register, GET /students/:id, PATCH /students/:id, savedPreceptorIds logic, input validation	Planned

Week 2 (Day 8 – Day 14) — Preceptor Module & Verification Workflow

Day	Hours	Task	Scope & Description	Status
Day 6	4 hrs	**Preceptor Profile APIs**	POST /preceptors/register, GET /preceptors/:id, PATCH /preceptors/:id — credentials, specialty, location, availableDays fields	Planned
Day 7	4 hrs	**License Verification Flow**	POST /preceptors/:id/submit-license, file upload to Firebase Storage, verificationStatus state machine (pending/approved/rejected)	Planned
Day 8	4 hrs	**Admin Verification APIs**	GET /admin/verifications, PATCH /admin/preceptors/:id/approve, PATCH /admin/preceptors/:id/reject, email notification triggers	Planned
Day 9	4 hrs	**Preceptor Search & Filter**	GET /preceptors with query params: specialty, location, availableDays, honorarium range — Firestore composite index optimization	Planned

Day 10	4 hrs	**Save / Bookmark Listings**	POST /students/:id/save-preceptor, DELETE /students/:id/save-preceptor/:pid, GET /students/:id/saved — savedPreceptorIds array management	Planned

Week 3 (Day 15 – Day 21) — Payments, Premium & Contact Reveal System

Day	Hours	Task	Scope & Description	Status
Day 11	4 hrs	**Stripe Integration Setup**	Stripe SDK config, product & price creation, webhook endpoint registration, PCI compliance headers, idempotency keys	Planned
Day 12	4 hrs	**Subscription Checkout API**	POST /payments/create-checkout-session, Stripe Checkout redirect, success/cancel URL handling, session metadata mapping	Planned
Day 13	4 hrs	**Webhook & Premium Activation**	POST /webhooks/stripe — handle checkout.session.completed, invoice.paid, customer.subscription.deleted; update isPremium flag in Firestore	Planned
Day 14	4 hrs	**Contact Reveal System**	POST /preceptors/:id/reveal-contact — isPremium gate check, create Preceptor_Contacts document, POST /notify trigger, audit log entry	Planned
Day 15	4 hrs	**Inquiry & Notification API**	POST /inquiries — create inquiry record, Firebase Cloud Function trigger, SendGrid/Nodemailer email dispatch to preceptor	Planned

Week 4 (Day 22 – Day 28) — Analytics, Admin Dashboard APIs & QA

Day	Hours	Task	Scope & Description	Status
Day 16	4 hrs	**Analytics Events Module**	POST /analytics/event — log profile_view, contact_reveal, inquiry_sent per preceptor; GET /preceptors/:id/stats aggregation	Planned
Day 17	4 hrs	**Admin Dashboard APIs**	GET /admin/stats — total users, premium count, monthly revenue, top preceptors; GET /admin/users, PATCH /admin/users/:id/status	Planned
Day 18	4 hrs	**Billing Portal & History**	GET /payments/history/:preceptorId, POST /payments/create-portal-session (Stripe Customer Portal), invoice download endpoint	Planned
Day 19	4 hrs	**Security Hardening & Testing**	Rate limiting (express-rate-limit), input sanitization, CORS policy, Helmet.js headers, JWT expiry refresh, Postman test suite	Planned

Day 20	4 hrs	**Deployment & Documentation**	Firebase Functions deploy, Firestore security rules finalize, API documentation (Swagger/OpenAPI), handover checklist	Planned

4. API Endpoints Master List

All 25 backend endpoints are listed below, organized by module. Each endpoint is secured via JWT authentication and RBAC middleware. Admin-only endpoints require an admin role claim in the token.

Module	Endpoint	Description
Auth	POST /auth/register	Register new user (Student or Preceptor)
Auth	POST /auth/login	Login and return JWT token
Auth	POST /auth/verify-email	Verify user email via token
Student	GET /students/:id	Get student profile
Student	PATCH /students/:id	Update student profile
Student	POST /students/:id/save-preceptor	Save a preceptor to bookmarks
Student	GET /students/:id/saved	Get all saved preceptors
Preceptor	POST /preceptors/register	Create preceptor profile
Preceptor	GET /preceptors	Search/filter preceptors (query params)
Preceptor	GET /preceptors/:id	Get preceptor public profile
Preceptor	PATCH /preceptors/:id	Update preceptor profile
Preceptor	POST /preceptors/:id/submit-license	Upload license for verification
Preceptor	POST /preceptors/:id/reveal-contact	Reveal contact (premium gate)
Inquiry	POST /inquiries	Send inquiry to preceptor (triggers email)
Payment	POST /payments/create-checkout-session	Initiate Stripe Checkout
Payment	POST /webhooks/stripe	Stripe event webhook handler
Payment	POST /payments/create-portal-session	Open Stripe billing portal
Payment	GET /payments/history/:id	Get payment history
Analytics	POST /analytics/event	Log analytics event
Analytics	GET /preceptors/:id/stats	Get preceptor performance stats
Admin	GET /admin/verifications	List pending verifications
Admin	PATCH /admin/preceptors/:id/approve	Approve preceptor verification

Admin	PATCH /admin/preceptors/:id/reject	Reject preceptor verification
Admin	GET /admin/stats	Platform-wide statistics
Admin	GET /admin/users	List and manage all users

5. Database Schema Summary

The Firestore data model is normalized around six core collections. Foreign key relationships are enforced via UID references and Firestore security rules.

Collection	Key Fields	Purpose
USERS	uid (PK), email, role, subscription, displayName	Master user record; role field drives RBAC
STUDENTS	uid (PK), university, program, savedPreceptorIds[]	Student profile & saved preceptor bookmarks
PRECEPTORS	uid (PK), specialty, isVerified, isPremium, licenseFileUrl	Preceptor profile, premium & verification state
PRECEPTOR_CONTACTS	id (PK), preceptorUid (FK), studentUid (FK), message	Audit log of all contact reveal events
PAYMENTS	id (PK), preceptorUid (FK), amount, status, createdAt	Stripe payment records for premium subscriptions
ANALYTICS_EVENTS	id (PK), preceptorUid (FK), type, ts	Tracks profile views, reveals, inquiries per preceptor

6. Risks & Mitigation Plan

Risk	Probability	Mitigation
Stripe webhook delivery failure	Medium	Idempotency keys + retry logic + Stripe Dashboard monitoring
Firestore read/write quota limits	Low	Composite indexes, batched writes, query optimization
License file upload size exceeded	Medium	Enforce 5MB limit client-side + server-side validation
Unauthorized contact reveal bypass	Low	Server-side isPremium check; never trust client flag
Email notification delays	Medium	SendGrid delivery tracking + Cloud Function retry policy
Scope creep delaying delivery	Medium	Strict 40-hour scope lock; new requests logged for v2

7. Final Deliverables

At the conclusion of the 40-hour development cycle, the following deliverables will be handed over:

- Fully functional REST API with 25 secured endpoints deployed on Firebase Cloud Functions
- Firestore database with all 6 collections, indexes, and security rules configured
- Stripe integration: Checkout, Webhooks, Customer Portal, Payment history
- Firebase Cloud Storage setup for preceptor license document uploads
- Email notification system via SendGrid/Nodemailer for inquiry and verification alerts
- Admin Dashboard APIs: user management, verification queue, platform KPIs
- Analytics event logging and aggregated preceptor performance stats
- Complete Swagger/OpenAPI 3.0 documentation for all endpoints
- Postman test collection with auth, happy-path, and edge-case test scenarios
- Firestore Security Rules file for production deployment
- README and environment setup guide for onboarding new developers

8. Sign-Off & Approval

This document is prepared for client review, validation, and contractual alignment. It defines the complete scope of backend engineering work for the NPaxis Platform one-month development cycle.

Role	Name	Signature / Date
Project Manager / Backend Lead	_____	_____
Client / Stakeholder	_____	_____
Technical Reviewer	_____	_____